Pally: Groupware Heuristics Evaluation

Session A is the baseline (Discord voice only); Session B uses Discord voice with our group bot.

Coding key

Present	The mechanism exists and worked during testing.
Constrained	Supported, but limited; users met difficulty or needed a workaround.
Absent	The task needed the mechanism, but the system did not provide it.
N/A	The test tasks did not trigger this situation, so it cannot be judged.

Evaluation

#	Heuristic	Session A Discord voice only	Session B With Pally	Evidence
1	**Provide the means for intentional and appropriate verbal communication**	Present	Present	Session A Discord voice supported real-time discussion, but participation was uneven. A and B often talked over or moved past C's ideas. During mode selection, C usually followed the others rather than insisting on their own preference. Session B C explicitly suggested using Pally as a way to express an opinion they did not feel confident raising directly. Pally therefore created an additional route into the discussion. C still introduced ideas cautiously, and when trying to leave, the tool worked less as a literal excuse and more as a low-pressure way to start saying “I want to leave.” This prompted A and B to acknowledge the intention and respond with “one last game.”
2	**Provide the means for intentional and appropriate gesture communication**	N/A	N/A	Session A The session used voice communication only. No video, pointing, or gesture-based task was involved. Session B Pally did not add gesture-based communication, and the test tasks did not require gesture or pointing. Therefore this heuristic could not be meaningfully assessed.

#	Heuristic	Session A Discord voice only	Session B With Pally	Evidence
3	**Provide consequential communication of an individual's embodiment**	Present	Present	Session A Discord showed who was in the voice channel and who was speaking, but these cues did not make C's unspoken preferences visible. A and B could hear C when C spoke, but often did not recognise C's quieter position in the discussion. Session B Discord continued to provide presence and speaking cues. Pally did not add richer embodiment information, but some previously hidden intentions became visible indirectly through tool use, such as C proposing Random Pick or using the leaving interaction.
4	**Provide consequential communication of shared artifacts**	N/A	Constrained	Session A No shared Pally artifact existed in the baseline session, so artifact feedthrough could not be assessed. Session B Pally's outputs became shared reference points. For example, once Random Pick selected C, A and B immediately recognised the result and were willing to accept C's choice. However, some contribution information remained hidden because tool actions and anonymous evaluations did not clearly show who submitted what. The group could see the outcome, but not always the individual action behind it.
5	**Provide protection**	Absent	Present	Session A The voice-only setting provided little protection for a quieter member who did not feel confident challenging the more vocal members. C often followed A and B's preferences instead of openly pushing a different opinion. Session B Pally provided a mediated and partly anonymous way to contribute. C proposed using the tool specifically when they did not feel comfortable stating the opinion directly. This reduced the exposure involved in disagreeing. However, protection did not automatically create equal recognition: during peer evaluation, A and B still tended not to comment on C's performance, whether positive or negative.
6	**Management of tightly and loosely-coupled collaboration**	Constrained	Present	Session A Interaction was tightly coupled through live voice discussion. To contribute, C had to enter the same real-time conversation dominated by A and B, which made quieter participation more difficult. Session B Pally introduced a looser contribution path alongside live voice. Participants could use a tool, submit input, and then return to the shared discussion. This gave C a way to surface an opinion without first competing for the floor. The group still moved back to live verbal negotiation afterwards, so the two modes worked together rather than replacing each other.

#	Heuristic	Session A Discord voice only	Session B With Pally	Evidence
7	**Allow people to coordinate their actions**	Constrained	Present	Session A Coordination depended mainly on spontaneous verbal negotiation. A and B tended to dominate decisions, while C often followed the preference already being pushed by others. Session B Random Pick changed the coordination pattern. When C was selected to decide, A and B, despite being the louder members, accepted C's choice much more easily. The tool gave C's decision temporary legitimacy. However, C still presented the choice cautiously and often handed the final confirmation back to A and B, showing that the existing power pattern was reduced but not completely removed. The leaving interaction also supported coordination around ending the session: once C's intention to leave was surfaced, the group shifted from ignoring that possibility to negotiating a shared transition, expressed as “one last game.”
8	**Facilitate finding collaborators and establishing contacts**	N/A	N/A	Session A All participants were already in the same Discord group and voice channel. Finding or inviting collaborators was not part of the test. Session B Pally's tested functions did not involve finding new collaborators or establishing new contacts, so this heuristic was not triggered.

Key Observations
Decision-making: Random Pick gave C a legitimate moment to decide. A and B accepted C's selection more readily than in the baseline, but C still behaved cautiously and deferred final confirmation to them.
Peer evaluation: A and B tended not to comment on C's performance, whether positive or negative. The tool did not automatically remove this imbalance in attention.
Leaving: The Excuse Generator functioned more as a playful opening for expressing the intention to leave than as a tool for inventing a reason. It did not necessarily let C leave immediately, but it made the intention visible and prompted the group to negotiate, for example by saying “one last game.”